

# ViREL-BUDGET: Reliability-Constrained Visual-Token Budgeting for Green Vision-Language Inference

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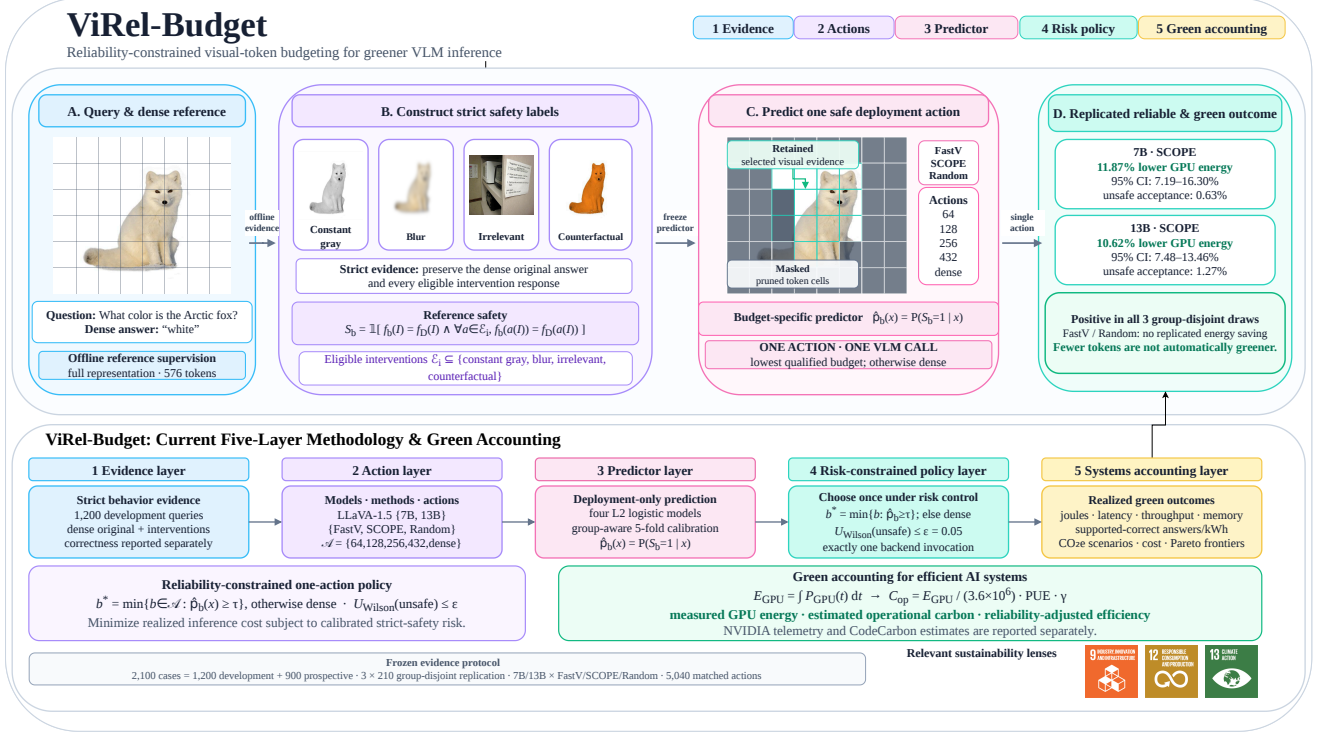


Figure 1: Overview of ViREL-BUDGET.

## Abstract

Visual-token pruning promises greener vision-language model (VLM) inference, but accuracy alone cannot show whether compression preserves visual reliance or produces real systems savings. We present ViREL-BUDGET, a reliability-constrained framework that constructs intervention-defined safety labels, learns budget-specific

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risk predictors without fine-tuning the VLM, and executes one selected action under a calibrated risk limit. We evaluate LLaVA-1.5-7B and 13B with FastV, SCOPE, and deterministic Random pruning using 1,200 development and 900 group-isolated prospective queries, and assess energy robustness across three independent 210-query draws. On the prospective evaluation, SCOPE satisfies the 5% risk limit with unsafe-acceptance rates of 1.67% at 7B and 2.89% at 13B. Across the separate replication draws, it reduces GPU energy by 11.87% and 10.62%, respectively. FastV and Random do not establish energy savings despite reducing tokens, showing that sparsity is not inherently green. Using paired GPU telemetry, carbon estimation, and reliability-adjusted efficiency metrics, ViREL-BUDGET identifies reliability-qualified configurations whose computational savings translate into defensible environmental benefits.

## CCS Concepts

• Information systems → Multimedia information systems; • Computing methodologies → Computer vision; • Hardware → Impact on the environment.

## Keywords

vision-language models, visual-token pruning, reliability-aware inference, visual reliance, adaptive computation, green AI, energy-efficient multimedia systems, carbon accounting

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## 1 Introduction

Vision–Language Models (VLMs) increasingly support multimedia search, reasoning, and interaction, yet representing images as hundreds of visual tokens increases inference latency, memory demand, energy use, and operational carbon. Visual-token pruning can reduce this burden while preserving average benchmark accuracy [4, 6]. However, such compression does not establish reliable green inference: an unchanged answer may conceal weakened dependence on visual evidence, and pruning overhead may prevent token reductions from lowering device energy. Hence, green VLM deployment must preserve image-dependent behavior and verify resource savings directly rather than infer them from token sparsity alone [8, 14].

We therefore introduce ViREL-BUDGET, a reliability-aware framework connecting visual-token pruning with deployment risk control and systems accounting. Dense and intervened-image evaluations provide development-only supervision for budget-specific behavior-preservation labels. Lightweight predictors use pre-inference information to estimate candidate safety, and a calibrated policy selects one pruned or dense action under a predeclared risk constraint. It executes the selected action once, without oracle evidence or additional VLM calls.

We investigate three research questions:

- **RQ1—Reliability:** Can a frozen controller select one pruned or dense action on unseen queries while satisfying a prospective unsafe-acceptance constraint?
- **RQ2—Variation:** How do reliability–resource trade-offs vary across evaluated LLaVA scales and pruning methods?
- **RQ3—Green systems outcomes:** Which reliable configurations reproduce measured energy savings across workload draws?

We evaluate six model–method controllers using 1,200 development cases and a group-isolated 900-query prospective set from POPE [11], MMStar [3], and Visual CounterFact [7]. All six satisfy the prospective reliability requirement; SCOPE records unsafe-acceptance rates of 1.67% at 7B and 2.89% at 13B. Across three independent 210-query replication draws, SCOPE reduces measured GPU energy by 11.87% at 7B (95% CI: 7.19%–16.30%) and 10.62% at 13B (95% CI: 7.48%–13.46%), whereas FastV and Random do not establish pooled energy savings despite avoiding dense inference. The framework separates task correctness from intervention-defined behavioral preservation and supports green claims only when reliability-qualified savings persist under direct measurement and repeated workload evaluation.

**Open Data and Code Statement.** To promote transparency, reproducibility, and reuse, we openly release the evaluation data and

accompanying documentation on Hugging Face<sup>1</sup>, together with the source code, configurations, and reproduction scripts on GitHub<sup>2</sup>.

## 2 Background and Related Work

### 2.1 Visual-Token Pruning

Vision-language models encode images as hundreds of visual tokens, increasing attention cost and inference latency. Visual-token pruning reduces this cost by retaining only a subset of the image representation. FastV uses attention observed in early transformer layers to remove low-importance tokens from subsequent computation, while SCOPE combines token saliency with coverage to preserve evidence distributed across the image [4, 6]. SparseVLM and DivPrune explore complementary text-guided and diversity-aware selection strategies [1, 18]. Although these methods can substantially reduce nominal token counts, their selection quality is not universal. Recent analyses find that learned importance heuristics can be competitive with or inferior to Random selection in some regimes, exposing the need for uninformed controls and budget-specific evaluation [16, 17]. More fundamentally, aggregate accuracy measures whether outputs remain correct on average, but not whether an individual pruned output preserves the evidence-dependent behavior of its dense reference. ViREL-BUDGET therefore treats pruning budgets as candidate deployment actions rather than assuming that a pruning score alone identifies a reliable output.

### 2.2 Visual Reliance

Answer correctness and visual reliance are not equivalent. A VLM may produce a correct or unchanged answer using language priors, memorized associations, object co-occurrence, or benchmark shortcuts even after relevant image evidence has been weakened. MM-Star addresses this problem through vision-indispensable questions, while POPE diagnoses object hallucination and Visual CounterFact introduces conflicts between visual attributes and learned priors [3, 7, 11]. VLind-Bench further measures the extent to which multimodal models favor linguistic priors over visual information [10]. These studies motivate controlled interventions that alter, degrade, or replace image evidence and observe whether the model’s response changes. Intervention sensitivity is an operational indicator of image-dependent inference, not proof that a model semantically understands the correct visual feature: response changes may also reflect distribution shift, uncertainty, or intervention artifacts. Accordingly, ViREL-BUDGET defines pruning safety through preservation of both the dense original answer and its responses under eligible interventions. These signals provide development supervision only; the frozen deployment policy must predict budget-specific risk without accessing dense answers, interventions, gold labels, or unexecuted-budget outcomes.

### 2.3 Green and Efficient AI Accounting

Green AI argues that predictive quality should be reported together with the computation, energy, and environmental cost required to obtain it [14, 15]. For pruning, token count and FLOPs are algorithmic proxies rather than direct measures of realized efficiency. Selection overhead, memory movement, kernel behavior, batching, hardware, and execution order can offset nominal reductions, so fewer visual tokens need not yield lower latency or device energy.

<sup>1</sup><https://huggingface.co/datasets/StableTradeAtlas/ViRel-Budget>

<sup>2</sup><https://github.com/StableTradeAtlas/ViRel-Budget>

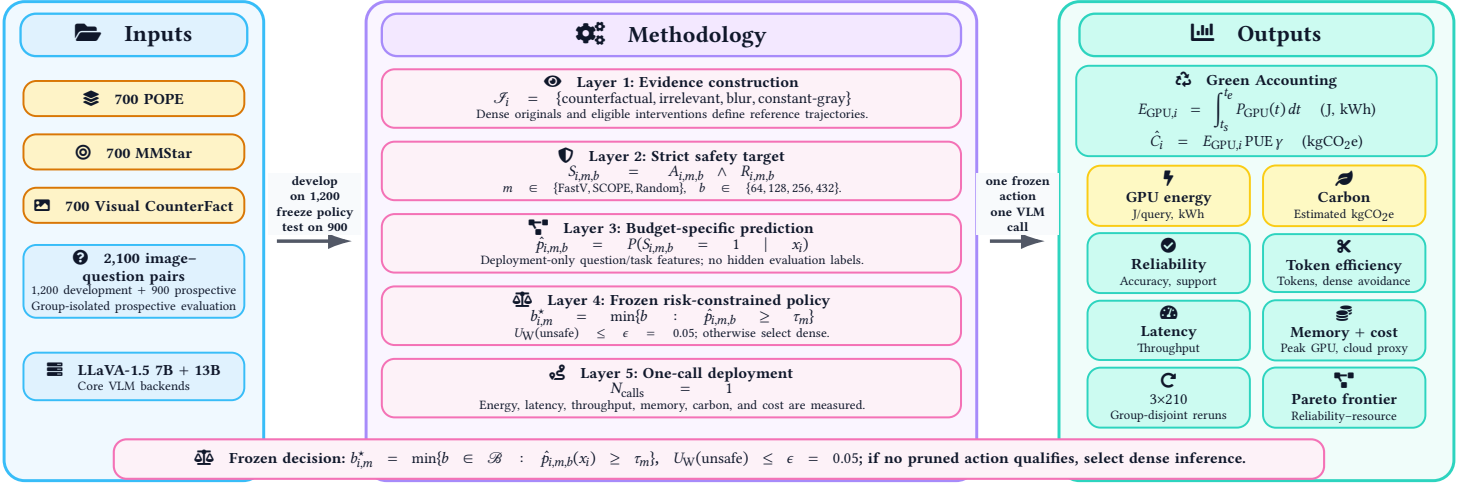


Figure 2: ViREL-BUDGET: intervention-guided, risk-constrained token budgeting for reliable and Green VLM inference.

Credible systems evaluation should disclose hardware and run-time, measure energy directly where feasible, quantify run-to-run uncertainty, and define the accounting boundary [8]. Estimating operational CO<sub>2</sub>e from energy requires explicit assumptions about grid carbon intensity and facility power usage effectiveness. Code-Carbon supports reproducible run-level estimation, while broader environmental tools distinguish operational estimates from embodied and lifecycle impacts [5, 13]. ViREL-BUDGET consequently evaluates one-action inference using measured GPU energy, latency, throughput, and memory alongside estimated carbon and cloud cost. Reliability-adjusted metrics and Pareto frontiers determine whether nominal sparsification produces a greener operating point without rewarding behaviorally unsafe outputs.

### 3 ViREL-BUDGET

#### 3.1 Overview and Evidence Construction

Given an image-question pair  $x_i = (I_i, q_i)$ , a vision-language model  $f_\theta$ , and a pruning method  $m$ , ViREL-BUDGET converts static token pruning into a reliability-aware deployment policy. It selects the smallest admissible budget while controlling the risk of changing the dense model’s original and intervention-conditioned responses, then evaluates the resulting resource savings. As shown in Figure 2, dense outputs and image interventions provide development-only supervision; the frozen policy selects an action before deployment inference.

Specifically, let

$$\mathcal{B} = \{64, 128, 256, 432, \text{dense}\}$$

denote the available actions, and let  $\Omega(b)$  represent their measured resource cost. The resulting decision problem is

$$\min_{\pi} \mathbb{E}[\Omega(\pi(x))] \quad \text{s.t.} \quad P(\text{unsafe acceptance}) \leq \epsilon, \quad (1)$$

where  $\pi$  is the budget policy and  $\epsilon = 0.05$  is the predeclared risk limit. Token count orders candidate actions during selection,

whereas energy, latency, and memory measurements together determine whether this algorithmic reduction produces an efficient or green operating point.

During development, the dense model is evaluated on the original image and a set  $\mathcal{J}_i$  of eligible interventions: targeted counterfactual replacement, irrelevant-image replacement, blur, and constant-gray input. An intervention is eligible when it changes the canonicalized dense answer, thereby defining an observable reference response rather than assuming that every transformation is meaningful. Hence, the resulting dense trajectory is

$$\mathbf{y}_i^D = (y_{i,0}^D, \{y_{i,j}^D\}_{j \in \mathcal{J}_i}), \quad (2)$$

where  $y_{i,0}^D$  is the original dense answer and  $y_{i,j}^D$  is its answer under intervention  $j$ .

For method  $m$  and budget  $b$ , original-answer fidelity is

$$A_{i,m,b} = \mathbf{1}\{y_{i,m,b,0} = y_{i,0}^D\}, \quad (3)$$

and intervention-trajectory fidelity is

$$R_{i,m,b} = \prod_{j \in \mathcal{J}_i} \mathbf{1}\{y_{i,m,b,j} = y_{i,j}^D\}. \quad (4)$$

The strict safety label and task-correctness label are kept separate:

$$\begin{aligned} S_{i,m,b} &= A_{i,m,b} \wedge R_{i,m,b}, \\ C_{i,m,b} &= \mathbf{1}\{y_{i,m,b,0} = y_i^{\text{gold}}\}. \end{aligned} \quad (5)$$

Thus, a safe action must preserve both the original dense answer and the eligible intervention responses, while task correctness is assessed separately. This separation provides an operational measure of visual-reliance preservation without conflating behavioral fidelity with predictive accuracy.

#### 3.2 Budget-Specific Prediction and Frozen Policy

Visual-token reliability is not strictly monotonic: retaining more tokens does not always better preserve the dense model’s response.

ViREL-BUDGET therefore predicts safety independently for each pruned budget:

$$\hat{p}_{i,m,b} = P(S_{i,m,b} = 1 \mid \mathbf{x}_i), \quad b \in \mathcal{B} \setminus \{\text{dense}\}. \quad (6)$$

Here,  $\mathbf{x}_i$  contains only pre-inference features: question word and character counts, option count, a multiple-choice indicator, one-hot dataset/task indicators, and predeclared keyword indicators for color, size, counting, existence, spatial relations, text, and emotion. Dense answers, intervention outputs, gold labels, and unexecuted-budget outcomes are excluded, preventing deployment-time access to supervisory evidence.

For each model-method pair, four  $L_2$ -regularized logistic regressions predict safety at the four pruned budgets. We use logistic regression as a lightweight instantiation of the risk-prediction layer, not as the framework’s principal algorithmic contribution; richer predictors may shift the reliability-coverage frontier. Model selection and calibration use group-aware five-fold cross-validation on the 1,200-case development pool, keeping source-related image-question instances in the same fold.

A threshold  $\tau_m$  is accepted only when the one-sided 95% Wilson upper bound on unsafe acceptance,  $U_W$ , does not exceed  $\epsilon$ . After cross-validation and threshold calibration, we refit the predictors on the full development pool and freeze the feature schema, thresholds, and fallback rule before prospective evaluation. Define the qualifying pruned actions as

$$\mathcal{Q}_{i,m} = \{b \in \mathcal{B} \setminus \{\text{dense}\} : \hat{p}_{i,m,b} \geq \tau_m\}. \quad (7)$$

The frozen decision is then

$$b_{i,m}^* = \begin{cases} \min \mathcal{Q}_{i,m}, & \mathcal{Q}_{i,m} \neq \emptyset, \\ \text{dense}, & U_W(\text{unsafe}; \tau_m) \leq \epsilon, \end{cases} \quad (8)$$

The minimum is taken according to retained visual tokens. Dense inference is selected if no pruned action qualifies. Unsafe acceptances remain visible in evaluation and are never repaired retrospectively; unnecessary dense fallbacks are counted as lost efficiency.

### 3.3 Single-Inference Deployment Protocol

The frozen policy makes one decision and executes exactly one VLM call,  $N_{\text{calls}}(\mathbf{x}_i) = 1$ . It does not try successively larger budgets, inspect a dense answer, or invoke post-hoc intervention tests. This distinction is central to the operational claim: a cascade may report the cost of its final answer while concealing the energy and latency of failed attempts. ViREL-BUDGET instead includes feature extraction, controller latency, the selected inference, and any dense fallback within the declared deployment boundary.

The same controller design is instantiated separately for LLaVA-1.5 7B and 13B [12] and for FastV [4], SCOPE [6], and deterministic Random pruning. Random pruning provides an uninformed control with the same token budgets, while FastV and SCOPE test whether structured visual-token selection produces better reliability-resource trade-offs. Dense inference remains the no-pruning reference.

The design of ViREL-BUDGET distinguishes three quantities: visual-token reduction, systems efficiency, and environmental benefit. Fewer retained tokens establish algorithmic sparsity, but a method is operationally efficient only when measured latency, throughput,

memory, or energy also improves. A green claim additionally requires that the resource saving survives the reliability constraint and the complete one-call execution boundary.

### 3.4 Green Evaluation, Accounting, Replication

Evaluation uses 2,100 image-question cases, balanced across MM-Star [3], POPE [11], and Visual CounterFact [7]. The original 1,200 cases form the development pool, while the remaining 900 cases are isolated by source group before inference and reserved for prospective evaluation. Controller decisions are recorded before hidden dense, intervention, and correctness labels are revealed.

For each executed action, we report accuracy, strict safety, unsafe acceptance, dense avoidance, retained tokens, latency, throughput, peak GPU memory, cloud-cost proxy, and backend calls. Green efficiency is evaluated using joules per query, joules per correct or intervention-supported correct answer, and correct or supported-correct answers per kWh. These reliability-adjusted measures prevent a low-energy but unreliable configuration from appearing environmentally preferable.

GPU energy is obtained by integrating synchronized NVIDIA power telemetry over the inference window. CodeCarbon 3.2.9 [5] records parallel estimates for the replication runs. Measured device energy and estimated carbon are distinguished as

$$E_{\text{GPU},i} = \int_{t_s}^{t_e} P_{\text{GPU}}(t) dt, \quad (9)$$

$$\hat{C}_i = E_{\text{GPU},i}^{(\text{kWh})} \times \text{PUE} \times \gamma,$$

where  $\gamma$  is the regional grid-carbon intensity. The central estimate uses  $\text{PUE} = 1.2$  and  $\gamma = 0.03283 \text{ kgCO}_2\text{e/kWh}$ ; sensitivity analysis spans  $\text{PUE} \in [1.1, 1.4]$  and  $\gamma \in [0.015, 0.080]$ . Following established Green AI reporting practice [8, 14], results identify the hardware, region, system boundary, measurement source, run count, and assumptions rather than merely reporting carbon as a scalar.

Three accounting scopes are maintained: deployment-workload energy, identifiable experimental energy, and a broader allocation-based footprint that includes recoverable setup or idle time. Historical runs with NVIDIA telemetry use direct GPU measurements; runs lacking telemetry are marked as reconstructed estimates. CodeCarbon is not used retrospectively to relabel unobserved runs.

Finally, because benchmark comparisons can vary with data sampling and other experimental factors [2], robustness is tested on three fixed, stratified, group-disjoint draws of 210 cases. Each draw uses matched queries and counterbalanced execution order across both model sizes and all pruning methods, with dense references measured under the same protocol. A configuration passes the predeclared strong-green replication gate (Gate R) only when it satisfies the reliability limit, reduces energy in every draw, and has a positive pooled lower confidence bound for energy saving. Reliability-energy and reliability-carbon Pareto frontiers are also constructed, so a configuration is preferred only when no alternative provides equal or better behavioral performance with lower resource use. This protocol connects efficient multimedia inference to reproducible and quantitatively supported green outcomes.

**Table 1: Prospective reliability and replicated energy change.**

| Model | Method | Prospective $n = 900$    |                              |                           | Rerun<br>$\Delta E$ (% , $\uparrow$ ) |
|-------|--------|--------------------------|------------------------------|---------------------------|---------------------------------------|
|       |        | Acc.<br>(%, $\uparrow$ ) | Unsafe<br>(%, $\downarrow$ ) | Avoid<br>(%, $\uparrow$ ) |                                       |
| 7B    | FastV  | 63.67                    | 1.78                         | 63.44                     | −3.11                                 |
|       | SCOPE  | <b>64.11</b>             | <b>1.67</b>                  | 34.33                     | <b>+11.87</b>                         |
|       | Random | 62.67                    | 3.44                         | 56.33                     | −5.96                                 |
| 13B   | FastV  | 72.56                    | 2.56                         | 77.56                     | −1.35                                 |
|       | SCOPE  | 72.22                    | 2.89                         | 50.00                     | <b>+10.62</b>                         |
|       | Random | 72.33                    | <b>2.00</b>                  | 51.78                     | −3.92                                 |

## 4 Results

We evaluate ViREL-BUDGET at three complementary levels. Prospective reliability is measured on 900 group-isolated queries after all six controllers are frozen. Energy replication uses three independently sampled, group-disjoint 210-query draws, yielding 630 unique queries. Budget-level safety and dense model-scale comparison use the 1,200-case development pool. We keep these populations separate throughout the section and report measured GPU energy separately from estimated carbon.

### 4.1 Prospective Reliability and Budget Behavior

All six frozen controllers satisfy the prospective risk requirement under the single-inference protocol defined in Section 3.3. Table 1 reports accuracy, unsafe acceptance, and dense avoidance on the sealed  $n = 900$  prospective population, together with the pooled energy effect from the separate replication population. The 7B controllers achieve accuracies between 62.67% and 64.11%, while the 13B controllers achieve between 72.22% and 72.56%. Unsafe acceptance ranges from 1.67% to 3.44% at 7B and from 2.00% to 2.89% at 13B. Specifically, SCOPE records 15/900 unsafe acceptances at 7B and 26/900 at 13B, corresponding to 1.67% and 2.89%, respectively. In every case, the one-sided 95% Wilson upper bound remains below the predeclared 5% limit.

The controllers occupy different reliability–coverage operating points. FastV avoids dense inference for 63.44% of prospective queries at 7B and 77.56% at 13B, the highest rates among the three pruning substrates. SCOPE is more conservative with dense-avoidance rates of 34.33% and 50.00% respectively. Random pruning avoids dense inference on 56.33% of 7B queries and 51.78% of 13B queries, but its prospective unsafe-acceptance rate reaches 3.44% at 7B. These results measure how often each frozen policy returns a pruned output within the common risk constraint; dense avoidance alone is not an efficiency outcome.

Budget-level results further separate pruning aggressiveness from behavioral safety. Figure 3(a) reports strict reference-safe rates at 64, 128, 256, and 432 visual tokens on the development pool. Population-level safety generally increases with budget, but the ordering among FastV, SCOPE, and Random changes across budgets and model scales. SCOPE has the highest strict-safe rate at the two most aggressive 7B budgets, whereas FastV reaches the highest rate at larger budgets and at most 13B operating points.

Figure 3(b) places the six prospective controllers on the resulting unsafe-acceptance–dense-avoidance frontier. All remain within the 5% risk boundary, while their coverage varies from 34.33% to 77.56%.

Finally, to identify what drives the budget predictors, we conducted a development-only grouped feature analysis using the original folds and risk-calibration procedure. Task/source indicators had the largest mean absolute standardized coefficient in all 24 budget-specific regressions and were the best individual feature family for dense avoidance in four of six controllers. Removing them reduced pooled out-of-fold dense avoidance by 18.33–32.08 percentage points in five controllers, with 13B FastV as the exception. Question-structure and semantic-keyword features made method- and scale-dependent contributions. No image-derived metadata are used.

### 4.2 Energy Savings and Environmental Footprint

SCOPE is the only evaluated pruning substrate that satisfies the strong energy-replication criterion at both model scales. The separate replication population comprises three stratified, group-disjoint draws of 210 queries each ( $n = 630$  pooled), with matched Dense, FastV, SCOPE, and Random executions in every draw. A configuration passes Gate R only when its energy reduction is positive in all three draws, the pooled paired-bootstrap 95% lower bound exceeds zero, and the unsafe-acceptance Wilson upper bound remains at most 5%.

On the pooled replication population, SCOPE records 4/630 unsafe acceptances at 7B and 8/630 at 13B, corresponding to 0.63% and 1.27%, respectively. It reduces matched call-window GPU energy by 11.87% at 7B (95% CI: 7.19%–16.30%) and 10.62% at 13B (95% CI: 7.48%–13.46%). Both effects are positive in every draw. Figure 4(a) reports the pooled estimates, while Figure 4(b) shows the draw-level effects.

FastV and Random do not reduce pooled call-window energy relative to matched Dense. The estimated changes are −3.11% for 7B FastV, −1.35% for 13B FastV, −5.96% for 7B Random, and −3.92% for 13B Random, where a negative reduction denotes higher energy use. The FastV intervals include zero, whereas both Random intervals remain on the higher-energy side of the dense reference. Thus, the high dense-avoidance rates in Table 1 do not consistently produce lower device energy.

The replicated SCOPE policies also reduce estimated operational carbon under the disclosed location and facility assumptions. Mean call-window energy decreases from 36.14 to 31.85 J/query at 7B and from 65.87 to 58.87 J/query at 13B. With a central PUE of 1.2 and electricity intensity of 0.03283 kgCO<sub>2</sub>e/kWh, these operating points correspond to approximately 0.349 and 0.644 gCO<sub>2</sub>e per 1,000 queries. These values are derived from measured GPU energy rather than direct emissions observations.

Figure 4(c) reports the broader experimental footprint separately from per-query deployment results. The recoverable ledger contains 66 runs, totaling 19.17 monitored GPU-hours and 4.8198 kWh of measured or recovered GPU energy. The central facility-adjusted estimate is 0.1899 kgCO<sub>2</sub>e; varying PUE from 1.1 to 1.4 and electricity intensity from 0.015 to 0.080 kgCO<sub>2</sub>e/kWh gives a range

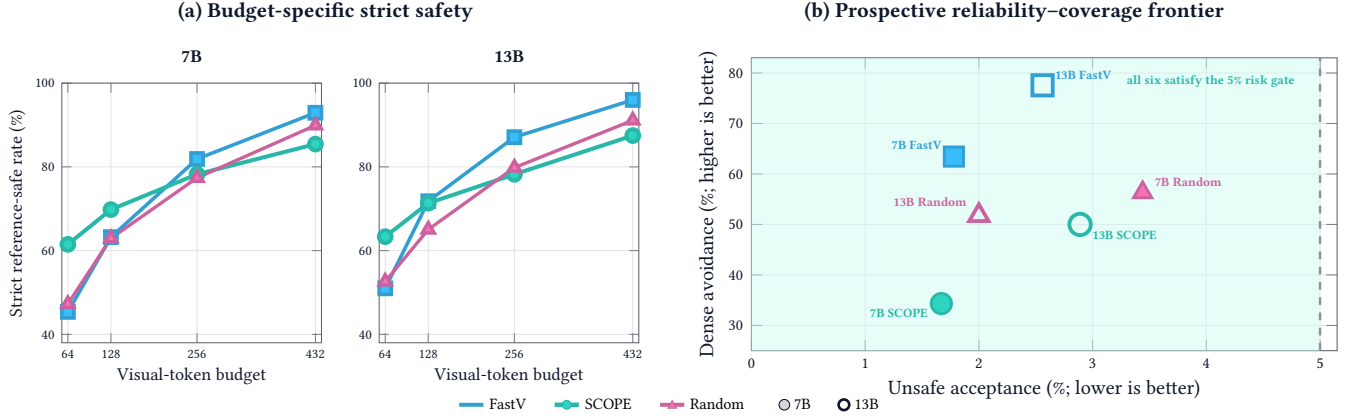


Figure 3: Strict safety increases with budget, and all prospective controllers remain below the 5% risk limit.

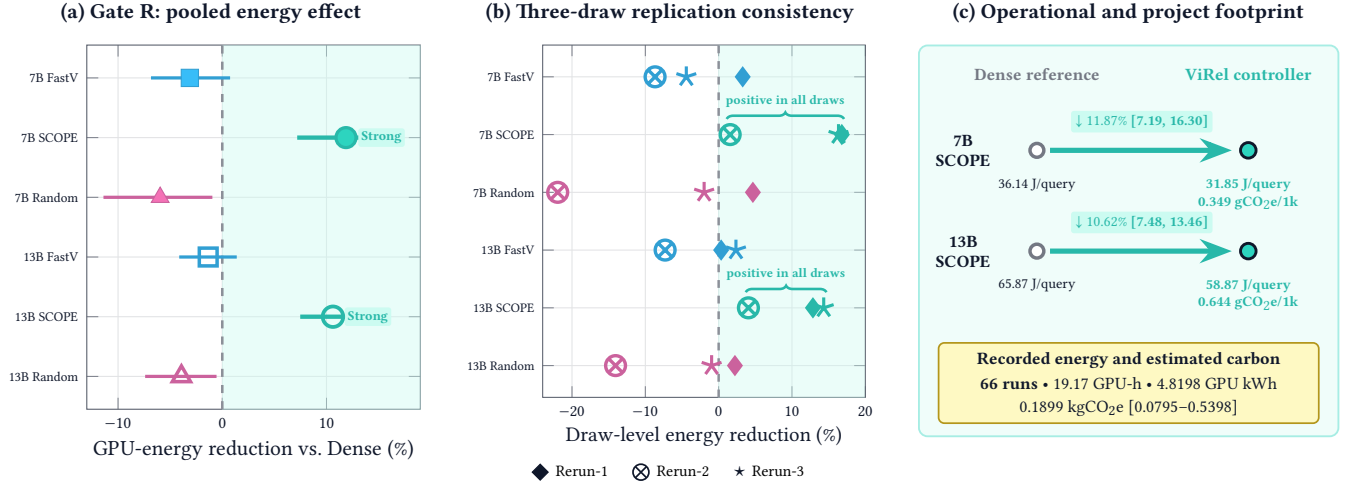


Figure 4: SCOPE consistently saves energy at both scales; project-level cost is reported separately.

of 0.0795–0.5398 kgCO<sub>2</sub>e. This boundary includes recorded development grids, prospective executions, replications, pilots, smoke tests, and recoverable failed attempts, but excludes unrecorded idle allocation and lifecycle impacts.

### 4.3 Efficiency and Model-Scale Pareto Frontiers

The validated operating points expose clear quality–resource trade-offs across model scales. On the 630-query replication population, 7B SCOPE achieves 66.51% accuracy, a 47.62% supported-correct rate, and 33.97% dense avoidance at 31.85 J/query. Its mean latency is 108.38 ms, corresponding to 9.23 queries/s and 53,820 supported-correct answers/kWh. The 13B policy raises accuracy to 72.86%, supported-correct performance to 54.44%, and dense avoidance to 50.00%, while using 58.87 J/query at 171.81 ms/query. Its throughput is 5.82 queries/s, with 33,291 supported-correct answers/kWh.

Figure 5(a) places all six controller configurations in the energy–accuracy plane. SCOPE supplies the lowest-energy reliability-qualified controller at each model scale. The 13B points occupy a higher-accuracy, higher-energy region, while the 7B points require less energy per query. Figure 5(c) reports the same scale trade-off across accuracy, supported-correct rate, dense avoidance, energy reduction, and throughput without combining them into a single score.

The dense model-scale comparison produces a different frontier. On the common 1,200-case development evaluation, SmolVLM-500M reaches 65.83% accuracy with 2.92 GB peak GPU memory, LLaVA-1.5-7B reaches 66.50% accuracy at 36.59 J/query and 122.65 ms/query, and LLaVA-1.5-13B reaches 74.00% accuracy at 66.33 J/query and 180.80 ms/query. As shown in Figure 5(b), SmolVLM provides the lowest memory demand, 7B provides the lowest measured energy and latency, and 13B provides the highest accuracy. No model dominates all quality and resource objectives.

Collectively, SCOPE is the only controller with replicated energy reductions at both LLaVA scales. The 7B configuration favors energy efficiency and throughput, 13B favors answer quality, and the dense SmolVLM reference minimizes memory, demonstrating that no operating point dominates every objective. Section 5 examines why token reduction does not consistently yield systems savings.

## 5 Discussion and Analysis

### 5.1 Reliability-Aware Token Budgeting

The central implication of our results is that visual-token budgeting should be treated as a reliability-constrained decision problem, rather than as the selection of a universally effective compression

ratio. Figure 3(a) shows that strict reference safety depends jointly on budget, pruning substrate, and model scale. Although safety generally improves as more tokens are retained, the relative ordering of FastV, SCOPE, and Random changes across operating points. Together with the observed case-level non-monotonicity, this behavior rules out a simple assumption that a larger budget necessarily repairs every pruning failure. A deployable system must instead estimate the risk of each candidate action and retain a dense fallback when no pruned action is sufficiently reliable.

The prospective evaluation confirms that all six frozen controllers satisfy the predeclared risk requirement without deployment-time oracle information or retrospective correction. The controller’s negligible CPU overhead further shows that reliability-aware selection introduces no material secondary computation.

Table 1 also reveals why dense avoidance cannot be interpreted independently of behavioral risk or systems cost. FastV produces the greatest prospective dense avoidance at both model scales, while SCOPE adopts a more conservative acceptance policy. This does not make either strategy universally preferable: aggressive acceptance increases the opportunity for savings but also exposes the system to more budget-specific errors, whereas conservative fallback protects reliability at the cost of coverage. ViREL-BUDGET makes this trade-off explicit and configurable. The pruning substrate determines which visual tokens survive; the framework determines whether the resulting computation is sufficiently reliable to accept.

This separation is the framework’s central methodological role relative to fixed-budget pruning. It is supported by three distinct substrates rather than by one favorable token-ranking heuristic. SCOPE provides the strongest reliability–efficiency outcome, FastV demonstrates that high pruning coverage can be obtained without a corresponding energy guarantee, and Random establishes an uninformed control. The framework is therefore not tied to the claim that one pruning rule dominates all others. Instead, it supplies a common evidence, risk, and accounting layer through which heterogeneous pruning mechanisms can be evaluated and deployed under the same behavioral contract.

## 5.2 When Visual-Token Sparsity Becomes Green Efficiency

Applying the reliability-qualified Gate R to the replicated energy results, SCOPE is the only evaluated pruning substrate that passes at both model scales. Its GPU-energy reduction is positive in every draw, and both pooled confidence intervals remain above zero, as shown in Figure 4(a–b). The replicated reductions of 11.87% at 7B and 10.62% at 13B are therefore not attributable to one favorable workload sample. Together with compliance with the frozen unsafe-acceptance limit, this draw-level replication supports the paper’s principal green claim.

The FastV and Random results further strengthen this conclusion. FastV avoids dense inference most frequently, yet its pooled energy intervals do not establish a reduction at either scale. Random similarly removes visual computation but consumes more call-window energy than its matched dense reference. These contrasts show that nominal sparsity can be offset by implementation and system overheads, and that indiscriminate token removal is not intrinsically energy efficient. They also demonstrate the selectivity of Gate R: the framework accepts a green claim for SCOPE while rejecting the

same claim for configurations that appear efficient only through token counts or a single execution. This is especially important for 13B FastV, whose positive effect in the initial prospective run did not persist across the three-draw replication. Accounting for workload variation prevents that isolated result from being promoted into an unsupported general conclusion [2].

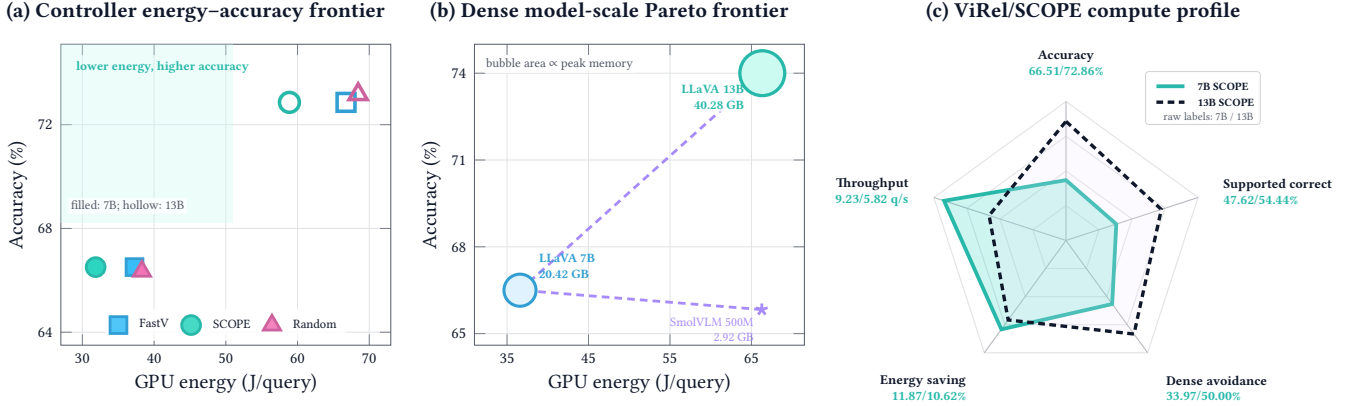
Environmental efficiency must likewise be normalized by useful behavior. Energy per query is informative, but it can favor a low-cost system that returns incorrect or intervention-unsupported answers. Metrics such as joules per supported-correct answer and supported-correct answers per kWh connect resource consumption to outputs that remain both task-correct and consistent with the defined visual-evidence behavior. On this basis, 7B SCOPE provides the strongest energy productivity among the validated SCOPE operating points, whereas 13B SCOPE exchanges part of that efficiency for higher accuracy and supported-correct performance. The appropriate objective is therefore not minimum energy in isolation, but minimum resource use for a required level of service and reliability.

Figure 4(c) extends this accountability from deployment to the research process itself. The recoverable experimental ledger reports measured GPU energy, monitored runtime, and estimated operational carbon for completed grids, pilots, replications, and recoverable failures. This project footprint must remain separate from the per-query benefit: it documents the environmental cost of producing the evidence, whereas the controller comparison estimates the cost of future deployment. Similarly, GPU energy is measured, while carbon is derived from disclosed PUE and regional electricity-intensity assumptions. The resulting values enable reproducible comparisons within the declared boundary, but they are not direct emissions observations or complete lifecycle estimates. Maintaining these distinctions avoids presenting carbon as a location-independent scalar and aligns the environmental claim with systematic reporting practice.

## 5.3 Pareto-Efficient Multimedia Inference and Deployment Implications

The model-scale results further show that responsible deployment cannot be reduced to choosing either the smallest or the most accurate model. Figure 5(a) places 7B SCOPE in the lower-energy region and 13B SCOPE in the higher-accuracy region. The 7B controller provides lower latency, greater throughput, and more supported-correct answers per kWh; the 13B controller provides higher accuracy, higher supported-correct performance, and greater dense avoidance. Both are rational operating points under different service requirements. Neither should be collapsed into a single “best” configuration without first specifying whether the deployment prioritizes quality, responsiveness, energy, or capacity.

The dense comparison in Figure 5(b) reinforces this conclusion. SmolVLM-500M has by far the lowest peak-memory demand and the greatest accuracy per parameter, but its tested implementation is neither the fastest nor the least energy-consuming. LLaVA-1.5-7B provides the lowest measured latency and energy, while 13B provides the highest accuracy. Thus, parameter count, token count, memory footprint, and device energy capture different aspects of efficiency. A smaller model can reduce memory requirements



**Figure 5: SCOPE defines the reliable energy frontier, while model scale governs the accuracy–memory–throughput trade-off.**

without automatically reducing latency or energy, just as a pruned model can reduce tokens without becoming greener.

These trade-offs motivate the pairwise Pareto analysis used in Figure 5, rather than an arbitrary scalar that merges accuracy, visual reliability, energy, latency, memory, and carbon. A configuration is useful when no alternative supplies equal or better behavioral performance with lower resource consumption for the deployment objective under consideration. Under this interpretation, SCOPE defines the replicated reliability-qualified energy frontier at both LLaVA scales, while the remaining configurations expose alternative coverage, memory, or accuracy trade-offs rather than a universal ranking.

The results establish ViREL-BUDGET as a control template for visual-token pruning within the evaluated LLaVA-1.5 family. Although its evidence, policy, and accounting layers are not architecturally specific, their transfer to other VLM backbones, modalities, and adaptive-computation mechanisms remains untested. Applying the framework to image resolution, early exits, quantization, or model routing is therefore a direction for future evaluation rather than a demonstrated result.

## 6 Conclusion, Limitations and Future Work

In conclusion, we introduce ViREL-BUDGET, a reliability-aware framework that selects a pruned or dense VLM inference action under a calibrated risk constraint. Our results support reducing computation only when behavioral reliability is maintained, and system-level savings are directly verified.

First, ViREL-BUDGET is designed to measure preservation of the dense model’s intervention-conditioned behavior, not to certify semantic grounding or causal use of the correct visual evidence. A targeted color replacement that changes the dense answer can provide useful evidence when the pruned model preserves both the original and counterfactual responses. By contrast, blur or constant-gray input may change an answer because of uncertainty, distribution shift, or degradation unrelated to the question; preserving that change need not demonstrate correct evidence use [9, 10]. The criterion also inherits errors from the dense reference. We therefore interpret intervention-defined safety as behavioral fidelity under declared eligible interventions, report task correctness separately, and leave question-conditioned, region-level, and human-validated interventions to future work. Second, the controller uses lightweight

question-structure, task-source, and semantic-keyword features. Our grouped development-set analysis identifies task/source indicators as the strongest consistent signal, although their contribution varies across model–method controllers and may limit transfer to unseen task families. The limited feature space can still produce unsafe acceptances or unnecessary dense fallbacks, and we do not compare alternative predictor classes. Future work will examine richer pre-inference features and stronger risk controls while including their extraction and probing costs within the operational boundary. The prospective results provide empirical risk evidence under the evaluated distribution, not a distribution-free guarantee under arbitrary deployment shift.

Realized efficiency also depends on implementation, hardware, and serving conditions. SCOPE’s energy reductions replicated at both model scales, whereas FastV and Random showed that dense avoidance does not necessarily reduce device energy. Evaluations across additional accelerators, batch sizes, and optimized kernels are needed to characterize hardware-specific reliability–energy frontiers.

Our controller experiments provide within-family evidence across two LLaVA-1.5 scales, two informed pruning methods, and a Random control. SmolVLM-500M serves only as a dense model-scale reference and does not establish cross-backbone transfer. The reported reliability and energy results are therefore specific to the evaluated implementations, workloads, hardware, and pruning substrates. Applying the framework to newer VLM families, including Qwen-VL and InternVL, or to document, video, edge, and embodied settings will require separate calibration and direct systems measurement.

Finally, GPU energy is measured within declared inference windows, whereas operational carbon is estimated using disclosed grid intensity and PUE. This boundary excludes embodied hardware emissions and incompletely observed networking, storage, and allocation overhead, and therefore is not a complete lifecycle assessment. Future work will incorporate full-system telemetry, temporally resolved grid intensity, lifecycle accounting, and carbon-aware scheduling [8, 15]. These extensions preserve the central requirement that efficiency claims remain reliability-qualified, boundary-explicit, and empirically verified.

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